

Name: _____

UHID: _____

Homework #03: Condition Codes

Condition Codes

Please predict the values of the listed condition flags after running the following instructions in each problem, which are assumed to run on a 32-bit ARM microprocessor in Thumb mode. If you think a flag cannot be determined, denote "X".
5pts each.

1.	MOVS R0, #-0x01 N: _____; Z: _____; C: _____; V: _____.
2.	MOV32 R1, #0xABAB MOV32 R2, #0x1212 ADDS R0, R1, R2 N: _____; Z: _____; C: _____; V: _____.
3.	MOV32 R1, #0x6000ABAB MOV32 R2, #0x60001212 ADDS R0, R1, R2 N: _____; Z: _____; C: _____; V: _____.
4.	MOV32 R1, #0x1000ABCD MOV32 R2, #0x8A00CDCD ADDS R0, R1, R2 N: _____; Z: _____; C: _____; V: _____.
5.	MOV32 R1, #0x1000ABCD MOV32 R2, #0xF000ABCD ADDS R0, R1, R2 N: _____; Z: _____; C: _____; V: _____.
6.	MOV32 R1, #0x80000000 MOV32 R2, #0x80000000 ADDS R0, R1, R2 N: _____; Z: _____; C: _____; V: _____.
7.	MOV32 R1, #0xFF00FF00

	MOV32 R2, #0xDD0DD00 ADDS R0, R1, R2 N: _____; Z: _____; C: _____; V: _____.
8.	MOV32 R1, #0x000000FF MOV32 R2, #-0xA0000000 ADDS R0, R1, R2 N: _____; Z: _____; C: _____; V: _____.
9.	MOV32 R1, #0x0000A0A0 MOV32 R2, #-0x000000FF ADDS R0, R1, R2 N: _____; Z: _____; C: _____; V: _____.
10.	MOV32 R1, #0xABAB MOV32 R2, #0x1212 SUBS R0, R1, R2 N: _____; Z: _____; C: _____; V: _____.
11.	MOV32 R1, #0x1212 MOV32 R2, #0xABAB SUBS R0, R1, R2 N: _____; Z: _____; C: _____; V: _____.
12.	MOV32 R1, #0x1000ABAB MOV32 R2, #0xFF001212 SUBS R0, R1, R2 N: _____; Z: _____; C: _____; V: _____.
13.	MOV32 R1, #0x1000ABAB MOV32 R2, #0x80001212 SUBS R0, R1, R2 N: _____; Z: _____; C: _____; V: _____.
14.	MOV32 R1, #0xFF000000 MOV32 R2, #0x22000000 SUBS R0, R1, R2 N: _____; Z: _____; C: _____; V: _____.
15.	MOV32 R1, #0xE0000000 MOV32 R2, #0x88000000 SUBS R0, R1, R2 N: _____; Z: _____; C: _____; V: _____.

16.	MOV32 R1, #0x8800FF00 MOV32 R2, #0xEE001100 SUBS R0, R1, R2 N: _____; Z: _____; C: _____; V: _____.
17.	MOV32 R1, #0x20000000 MOV32 R2, #-0x90000000 SUBS R0, R1, R2 N: _____; Z: _____; C: _____; V: _____.
18.	MOV32 R1, #0x0000A0A0 MOV32 R2, #-0x000000FF SUBS R0, R1, R2 N: _____; Z: _____; C: _____; V: _____.
19.	MOV32 R1, #0x0000A0A0 MOV32 R2, #0x00000100 MULS R2, R1, R2 N: _____; Z: _____; C: _____; V: _____.
20.	MOV32 R1, #0x40000000 MOV32 R2, #0x00000003 MULS R2, R1, R2 N: _____; Z: _____; C: _____; V: _____.